

DNA, it loses its double helix structure. Senior author and professor Alexis Vallée-Bélisle explains that, biochemicals such as proteins or RNAs act as nano-temperature-sensors inside a living organism and exhibit corresponding movements by folding or unfolding. Inspired by such natural phenomena, scientists have tried to incorporate similar technology using DNA where it folds and unfolds at predefined temperatures.

Prof. Vallée-Bélisle explained that their research breakthrough is an initial step taken to realise the complexities of thermo-regulation system at nano-scale. As an example, he cited that human body is maintained at a temperature of 37°C but till now, researchers have little idea regarding how an individual cell contributes to maintain that.

Knocki can make any surface smart enough to respond to taps

Knocki is a small white wireless device that can sit or be mounted on any smooth surface to transform it into a smart-remote for controlling home devices. Makers of Knocki have invented a patent-pending technology called SurfaceLink that is installed on Knocki to pick up taps that you make on the installed surface whether it is a wall, table top or door.



Knocki is a smartdevice that transforms walls, tables, doors and other surfaces into powerful remote controls (Image courtesy: www.geeky-gadgets.com)

To get things started, you have to fire up the companion Android or iOS app and configure the tap sequence to do a particular task. Tap sequences can range from simple single taps to multiple taps to a combination of taps. Then, you can assign a tap sequence to do a particular task.

Knocki also serves as a security device. If you have it installed on your door and paired it with your home security system, you will be able to knock your way into your home by tapping the set password knock sequence. You can activate your security alarm and know when the door is opened. You can use Knocki to send pre-programmed alerts or texts to your family members to let them know that you are leaving or have reached home.

New battery has a lengthened lifespan

Researchers at University of California, Irvine, USA, have invented nano-wire based battery material that can be recharged thousands of times, moving us closer to a battery that would never require replacement. This breakthrough could lead to commercial batteries with greatly lengthened lifespans for computers, smartphones, appliances, cars and spacecraft.

Scientists have long sought to use nano-wires in batteries. The researchers have solved this problem by coating a gold nano-wire in a manganese-dioxide shell and encasing the assembly in an electrolyte made of a Plexiglas-like gel. The combination is reliable and resistant to failure.

The study leader, UCI doctoral candidate Mya Le Thai, cycled the testing electrode up to 200,000 times over three months without detecting any loss of capacity or power and without fracturing any nano-wire.

Peggy, the smart clothes peg

Unilever's laundry detergent brand OMO has unveiled the world's first connected clothes peg named Peggy in the Australian market. On top of Peggy there is a UV sensor,



Peggy warns users when the washing is dry and when it is about to rain (Image Courtesy: <https://www.omo.com.au/peggy>)

which is used to monitor the amount of sunlight. This helps the connected app to tell the user if the conditions are suitable for drying clothes. This same information is cemented with the help of a temperature sensor and a humidity sensor. Both sensors also let the user know if the weather outside is being dominated by rain clouds or not. The connected app also lets you know if rains are imminent and whether you should collect the clothes from outside.

Peggy is being marketed with OMO, and so the connected app also features tools that help users with wash cycles. For example, the app tells which time is best suitable for washing your clothes. Depending upon the wash load, it calculates the time that will be taken to complete the wash and dry cycles.